Deterministic Subquadratic Reporting for Two-Dimensional Range LIS

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Abstract

Given a sequence of n real values and m rectangles in index–value space, we give a uniform deterministic algorithm reporting a longest strictly increasing subsequence in every rectangle in $O((n+m)n^{2/3}\log^{30}(n+m+2)) + O(K)$ operations, where K is the total output length. The result allows repeated values, arbitrary overlaps and all answer lengths. It answers the deterministic subquadratic existence question posed by Karthik C. S. and Saladi Rahul for uncolored two-dimensional reporting, while remaining slower than their randomized square-root bound.

The main step implements deterministic relative sampling for an implicit family of canonical longest chains. A trace on selected points must be computed through unselected original points; recomputing a longest chain on the selected points gives the wrong range space. We compute exact compressed source trees on original index intervals and charge their construction over disjoint intervals in merge/reduce. Signed descendant sums evaluate every conditional score, and upward-rounded integer probability tables give a finite-word deterministic compression. The resulting hitting set combines with truncated-LIS and endpoint-stitching routines, expanding only final witnesses. Generic relative sampling, merge/reduce, conditional expectation and the reporting framework are established ingredients; the contribution is their background-sensitive implementation and full charged integration.

1 The problem and the algorithmic advance

For real values $a_1, \dots, a_n$, identify position i with the point (i, a_i). A query $Q = [\ell, r] \times [a, b]$ asks for a longest sequence $i_1 < \cdots < i_h$ of points in Q with $a_{i_1} < \cdots < a_{i_h}$. The output consists of the positions themselves. Let K be the sum of the reported lengths. Input-value comparisons and operations on $\Theta(\log(n + m + 2))$-bit words have unit cost; arithmetic on more words is charged. No arithmetic on the real values is needed.

Theorem 1. *For every $n \geq 1$, every $m \geq 0$, every real sequence of length n and every family of m index–value rectangles, a uniform deterministic algorithm reports an optimal strict increasing subsequence in each rectangle using*

$$O\big((n + m)n^{2/3} \log^{30}(n + m + 2)\big) + O(K)$$

comparison and word operations. The constants are universal.

In particular, $m = \Theta(n)$ gives truly subquadratic overhead. Empty answers, ties, arbitrary overlaps and every optimum length are included. The exponent 30 is a conservative envelope for the explicit costs below. Karthik C. S. and Saladi Rahul [KR26], Theorems 1 and 5 and Section 7 of the full version [KR25], give randomized square-root-overhead two-dimensional reporting, a deterministic truncated-LIS routine, and the deterministic subquadratic question. We retain their short/long decomposition: solve short answers directly and hit a canonical optimum of every long query with a small stitching set. Our exponent is larger than their randomized exponent; the advance is a deterministic full-model algorithm.

The sampling rate itself is established. Relative approximation bounds and deterministic constructions appear in [HPS11], Theorem 2.11 and Lemma 2.15 of its preprint. Deterministic merge/reduce and subsystem oracles appear in [BCEG04], Section 2. For canonical chains, the selected trace depends on original points outside the selected set. Thus those generic results do not provide a free subset-only oracle. We implement an oracle costing $O(|I|s \log^2 n + s^2)$ for s selected occurrences in an original index interval I. Disjoint original intervals at each merge level make that extra background cost affordable. Compressed-tree score aggregation and finite-word compression complete the implementation.

Consistent-path covers and explicit-family path hitting are prior tools [TSP11, ACC19]; neither small-cover existence nor conditional expectation is claimed as a separate contribution. Strong deterministic index-interval LIS results [GGK25, Kor26] concern subarrays, without an independent value interval filtering every internal point. We make no improvement claim for that one-dimensional interface. The bounded literature comparison does not certify exhaustive priority.

The research and exposition used substantial OpenAI Codex assistance for proof development, literature comparison, exact diagnostic code and writing; the final methodology statement specifies its scope. The finite computations are diagnostics, and the proof below supplies the universal argument.

2 Canonical paths and the full-background trace oracle

Write $u \prec v$ if $u < v$ and $a_u < a_v$. For comparable endpoints let $P(u, v)$ be a longest chain from u to v, breaking ties by the lexicographically least sequence of original indices. Include singletons $P(u, u)$ and the empty chain; call this global family $\mathcal{F}$.

Lemma 2. *Every subpath of $P(u, v)$ between vertices x, y is $P(x, y)$. For a multiset S of s selected occurrences, every nonempty trace of $\mathcal{F}$ on S is the trace of $P(x, y)$ for two selected vertices. There are at most d^2 such descriptors, where $d \leq s$ counts distinct real vertices.*

Proof. Replacing a subpath by a longer chain improves the entire length; replacing it by an equal-length lexicographically earlier chain improves its tie order. Each replacement stays feasible because its internal points lie strictly between the two endpoints in both coordinates. This proves consistency. For a nonempty

trace take its first and last distinct selected vertices x, y. Consistency gives the asserted trace. All occurrences of a repeated vertex belong together. Dummy occurrences belong to no path. Empty traces remain empty under subsequent draws and require no descriptor. □

A rectangle optimum with endpoints u, v can be replaced by $P(u, v)$: every point of that path lies in the endpoints' coordinate box, hence in the query. The replacement has at least the optimum length and is therefore optimal. Hitting all members of $\mathcal{F}$ of length at least t suffices for every query with optimum at least t.

For context this family has VC-dimension at most two. If three index-ordered points x, y, z lie on a canonical path, every canonical path containing x, z contains $P(x, z)$ and hence y. The triple and its $\{x, z\}$ subset cannot both be traces. If no path contains the triple, it cannot be shattered either. This standard observation gives no efficient oracle by itself.

Suppose the real selected vertices lie in a contiguous original index interval I. Every feasible path between them stays inside I. Computing canonical paths using *all original points of* I therefore gives the global paths. Computing them on the selected set alone is wrong: for values $(0, 3, 1, 2, 4)$, $P(1, 5) = (1, 3, 4, 5)$ has trace $\{1, 5\}$ on $\{1, 2, 5\}$, whereas the selected-only longest path is $(1, 2, 5)$.

Lemma 3. *All trace descriptors on S can be represented by compressed source trees, with occurrence counts and constant-time membership tests, in $O(|I|s \log^2(n+2) + s^2)$ operations and $O(|I| \log(n+2) + s^2)$ working words.*

Proof. Give equal real values equal ranks by comparison sorting. For each distinct selected source u, sweep original positions from u rightward in I. Initially only u is reachable. A Fenwick structure queries ranks strictly smaller than the current rank and chooses a predecessor with maximum source-path length, then lexicographically least source path. Append the current vertex to the chosen predecessor. Induction over the sweep computes $P(u, v)$ for every reachable v: all feasible predecessor choices are considered, and appending a common endpoint preserves equal-length order.

The predecessor pointers form a rooted tree. A parent always precedes its child, so maintain binary lifting as vertices are inserted. To compare two equal-length root paths, find their lowest common ancestor and compare the indices of the children immediately below it. These are the first unequal entries. This costs $O(\log(n+2))$ per comparison, making the sweep cost $O(|I| \log^2(n+2))$ per source, including initialization and sorting. No path scans, probabilistic hashes or unbounded lexicographic integers occur.

Mark selected vertices. In index order record the nearest selected ancestor of each reachable original vertex, including itself when selected. For a selected $v \neq u$, its compressed parent is the nearest selected ancestor of its original parent. The root-to-v path in this compressed tree is exactly the selected trace of $P(u, v)$. Discard the original source tree after compression; keep one at-most-d-vertex tree per selected source. Each reachable pair (u, v) is a descriptor, including (u, u).

If w_x is the occurrence multiplicity of selected vertex x, a traversal adds w_x along each compressed root path, giving its input occurrence count. Euler entry/exit labels test whether a selected x is an ancestor of the descriptor target in constant time; this is exactly trace membership. A dummy is never a member. The compressed arrays have $O(s^2)$ entries, and all original-tree work is charged to the $|I|$-dependent term. □

3 Relative compression with rounded conditional probabilities

Consider $s = 2k$ input occurrences and $M \leq d^2 \leq 4k^2$ nonempty descriptors from Lemma 3. Let $p, \eta \in (0, 1]$ be rational. For a trace with a input occurrences put $u = a/(2k)$. A sequence of k uniform independent occurrence draws is bad for this trace if

$$\left| \frac{c}{k} - u \right| > \eta(p + u), \tag{1}$$

where c is its output count. The tolerance uses the *input sample* frequency, not an unavailable original frequency.

LEMMA 4. *If $2M\exp(-k\eta^2 p/3) \le 1/8$, a deterministic compression chooses k occurrences so that no descriptor is bad. In addition to constructing the source trees, it uses $O(k^3\,\mathrm{polylog}(n+2))$ word operations when $k < N$, where $N = O(n)$ and all parameter numerators and denominators have $O(\log(n+2))$ bits up to fixed factors.*

PROOF. For the mean of k Bernoulli variables of mean u, the two-tail bound is

$$\Pr[|c/k - u| > x] \le 2\exp\left(-\frac{kx^2}{2u + 2x/3}\right).$$

Indeed $\log \mathbb{E}e^{\lambda(X-u)} \le u(e^\lambda - 1 - \lambda) \le u\lambda^2/(2(1-\lambda/3))$ for $0 \le \lambda < 3$. Markov's inequality with $\lambda = x/(u+x/3)$ gives the upper tail. The lower tail follows from $e^{-\lambda} - 1 + \lambda \le \lambda^2/2$, with denominator $2u$; it is empty for $x > u$. Means zero and one are deterministic. Substituting $x = \eta(p+u)$ and $2u + 2x/3 \le (8/3)(p+u)$ gives the relaxed bound $2e^{-k\eta^2 p/3}$. Thus the initial expected number of bad descriptors is at most $1/8$.

For $0 \le a \le 2k$, $0 \le h, c$ with $c + h \le k$, let $q[a,h,c]$ be the exact probability of (1) after h further draws. Its terminal value is the bad-event indicator and

$$q[a,h,c] = \frac{aq[a,h-1,c+1] + (2k-a)q[a,h-1,c]}{2k}.$$

Choose a power of two $D \ge 64Mk^2$ and construct the integer table

$$\begin{aligned} U[a,0,c] &= D\mathbf{1}\{(1)\text{ holds}\}, \\ U[a,h,c] &= \left\lceil \frac{aU[a,h-1,c+1] + (2k-a)U[a,h-1,c]}{2k} \right\rceil. \end{aligned} \tag{2}$$

Convex averaging and one rounding unit per level prove by induction that

$$q[a,h,c] \le U[a,h,c]/D \le q[a,h,c] + h/D. \tag{3}$$

At a prefix with h draws remaining, maintain count c_F for each trace. Choose an occurrence x minimizing the integer score

$$V_x = \sum_F U[a_F, h-1, c_F + \mathbf{1}\{x \in F\}], \tag{4}$$

breaking ties by original occurrence position, and update all counts. Let Q be the true sum of conditional bad probabilities. Averaged over the $2k$ occurrences, the true next sums equal Q, counting multiplicities. By (3) the mean next upper score divided by D is at most $Q + M(h-1)/D$. The minimum score is at most that mean, and bounds the true chosen next sum from above. The total possible increase in true potential is therefore at most

$$\sum_{h=1}^{k} \frac{M(h-1)}{D} = \frac{Mk(k-1)}{2D} \le \frac{1}{128}.$$

The terminal potential is the integer number of bad descriptors and is less than $1/8 + 1/128 < 1$. It is zero. Exact monotonicity of the rounded scores is unnecessary. If $M = 0$, all occurrences are dummy and k dummy draws suffice without a table.

It remains to compute *every* candidate score efficiently. At one draw write $b_F = U[a_F, h-1, c_F]$, $\delta_F = U[a_F, h-1, c_F+1] - b_F$, and $B = \sum_F b_F$. Then $V_x = B + \sum_{F \ni x} \delta_F$. For each selected-source tree add $\delta_{(u,v)}$ at target v and sum accumulators from leaves to root. The accumulator at x is exactly the sum over descendant targets, hence over root-to-target traces containing x. Sum these values over sources. A real occurrence has this sum plus B; a dummy has score B. Differences may be negative and are stored with signs. Repeated

occurrences share a score but retain their multiplicities in averaging and in the output. All descriptors, traversals, candidates and Euler-membership count updates take $O(k^2)$ operations per draw, hence $O(k^3)$ in total. The table also has $O(k^3)$ entries, including all values of a.

For $k < N$ we have $M < 4N^2$. Take the least power of two at least $64Mk^2$, so $D < 128Mk^2$ and $\log D = O(\log N)$. Equation (3) gives $0 \le U \le D + k$. Products in (2) and signed sums of magnitude at most $O(M(D + k))$ use a constant number of words. In our application k is a power of two, so division by $2k$ is a shift; even elementary polylogarithmic arithmetic suffices. All counters, table entries and updates are charged, and no exact probability oracle or large-binomial unit-cost operation is used. □

4 A deterministically constructed stitching set

For $n \ge 2$, let N be the least power of two at least n, padding with $N - n$ dummy positions. Set

$$\begin{aligned} t &= \min\{z \in \mathbb{Z}_{\ge 1} : z^3 \ge n^2\}, \qquad L = \lceil \log_2(N+2) \rceil, \qquad H = L + 1, \\ p &= t/N, \qquad \eta = 1/(16H), \qquad k = 2^{\lceil \log_2(512 p^{-1} \eta^{-2} L) \rceil}. \end{aligned} \tag{5}$$

These are integer/rational calculations: find t by binary search and cube comparisons, and k by doubling and exact comparison. Thus $t = \Theta(n^{2/3})$ and $k = O(n^{1/3} \log^3(n+2))$. Parameter calculations use $O(\log(n+2))$ bits up to fixed constants. For these parameters the terminal bad-event test in (2) is precisely

$$16HN|2c - a| > 2kt + aN,$$

so it too is an integer comparison of the stated bit length.

If $k \ge N$, use every real point as the stitching set and skip compression. We charge this branch in Section 6. Otherwise k divides N. Partition padded index order into consecutive k-position blocks, represented initially by their exact occurrences. Merge adjacent equal-size *original* intervals, joining their k representatives into $2k$ occurrences and compressing to k with the full original parent interval as background. The children have both equal original sizes and equal representative sizes.

At every merge $M \le 4k^2 < 4N^2$ and $k\eta^2 p \ge 512L$. Since $e^L > 2^L \ge N + 2$, we have

$$2Me^{-k\eta^2 p/3} \le 8N^2 e^{-(512/3)L} < 1/8.$$

Lemma 4 therefore applies uniformly, independently of the representatives and their multiplicities. Lemmas 3 and 4 give per-merge cost

$$O\big(|I|k \log^2(n+2) + k^3 \operatorname{polylog}(n+2)\big). \tag{6}$$

Lemma 5. *The construction returns a set R of at most k real points hitting every canonical path of length at least t. On the branch $k < N$, its construction cost is $O(Nk \log^3(n+2) + Nk^2 \log^2(n+2))$.*

Proof. Fix any $F \in \mathcal{F}$. At a node let ϕ be its frequency among original padded positions and r its representative frequency. After j compression levels we prove simultaneously for all F that

$$|r - \phi| \le E_j(\phi + p), \qquad E_j = (1+\eta)^j - 1. \tag{7}$$

The base blocks are exact. At a merge both original and representative frequencies average their two children with equal weights. If u is the joined representative frequency, $|u - \phi| \le E_j(\phi + p)$. Lemma 2 and the full-background oracle cover its trace; an empty trace stays exactly empty. Compression adds at most $\eta(p + u)$. Since $p + u \le (1 + E_j)(p + \phi)$, the new error is at most $[E_j + \eta(1 + E_j)](\phi + p) = E_{j+1}(\phi + p)$. This proves the invariant, including repeated representatives and dummy occurrences.

There are at most H levels. Bounding binomial coefficients by H^i gives $(1 + 1/(16H))^H < \sum_{i \ge 0} 16^{-i} = 16/15$, so $E_H < 1/15$. At the root, a path of length at least t has $\phi \ge p$ and hence $r \ge \phi - (\phi + p)/15 \ge 13\phi/15 > 0$. Discard dummies and deduplicate only at the end. This preserves every hit and gives $|R| \le k$.

For cost, parent original intervals are disjoint at each level, so the background term of (6) sums to $O(Nk\log^2(n+2))$ per level and $O(Nk\log^3(n+2))$ overall. There are $N/k-1<N/k$ merge nodes. The k^3 term sums to $O(Nk^2\log^2(n+2))$, where a squared logarithm more than covers elementary arithmetic on a constant number of words. Original source trees are processed one at a time. Working memory is bounded by these time charges. In particular there is no global-n scan at every small merge and no enumeration of all global endpoint paths. □

Substituting (5) gives construction time $O(n^{5/3}\log^8(n+2))$. The sample scale follows the established relative sampling principle; the lemma supplies the additional original-background implementation and its global accounting. The constants are conservative and make no claim of practical efficiency.

5 Reporting all lengths with additive output cost

We give the details of the truncated-LIS and stitching integration to keep the full reporting guarantee explicit. Their basic framework is from [KR26, KR25]. All intermediate candidates will be feasible, and only the final winner for a query is expanded.

5.1 Short answers

Build a balanced binary tree on the original index order. At each split into children I_L, I_R, for $v\in I_L$ and $1\le\alpha\le t$ store $A(v,\alpha)$, the minimum last value of an α-chain starting at v within I_L. For $v\in I_R$ store $B(v,\beta)$, the maximum first value of a β-chain ending at v within I_R. Infeasible states have separate flags, denoted $+\infty$ and $-\infty$ respectively. Length-one states equal a_v. For larger lengths the recurrences are

$$A(v,\alpha)=\min_{\substack{w>v,\ a_w>a_v\\ w\in I_L}} A(w,\alpha-1), \tag{8}$$

$$B(v,\beta)=\max_{\substack{w<v,\ a_w<a_v\\ w\in I_R}} B(w,\beta-1). \tag{9}$$

Separating the first or last element proves these formulas. Store an attaining link to a previous-length state, giving an output-linear witness. For each length, strict-rank Fenwick sweeps cost $O(|I|\log(n+2))$. Summing over all splits and levels costs $O(nt\log^2(n+2))$ and stores $O(nt\log(n+2))$ states; cap lengths by child size.

For each split and length, store feasible A records in a static three-dimensional range-min structure with coordinates $(v,a_v,A(v,\alpha))$, weight $A(v,\alpha)$ and a witness handle. Use range-max structures for B records. Nested balanced range trees provide $O(\log^3(n+2))$ query time and, conservatively, $O(q\log^3(n+2))$ preprocessing on q records: recursively decompose a box into canonical subsets and store an extremal record at each final subset. This gives $O(nt\log^4(n+2))$ total preprocessing, using only coordinate comparisons and stored finite extrema.

Normalize query endpoints by comparison/binary search. Empty and single-position queries are direct. For $\ell<r$ let the lowest index-tree node containing both endpoints split at s, so $\ell\le s<r$. For each positive length up to t obtain

$$L_\alpha=\min\{A(v,\alpha):\ell\le v\le s,\ a\le a_v\le b,\ A(v,\alpha)\le b\},$$
$$R_\beta=\max\{B(v,\beta):s<v\le r,\ a\le a_v\le b,\ B(v,\beta)\ge a\}.$$

Every witness lies inside the query: internal indices and values lie between its endpoints. Conversely any feasible left α-chain has last value at least the A value of its start, and that minimizing witness satisfies the same boundaries. Thus L_α is exactly the minimum possible left peak. Similarly R_β is the maximum possible

right base. Two positive witnesses concatenate precisely when $L_\alpha < R_\beta$; query membership uses inclusive boundaries, but this join is strict.

Include zero-length empty states and every feasible one-sided chain. The feasible L_α are nondecreasing with α: drop the last element of any longer chain. Similarly R_β are nonincreasing, by dropping the first element. Infeasible lengths form suffixes. For each $\alpha \le t$ binary search the largest compatible $\beta \le t$ and retain the maximum $\alpha + \beta$, also considering $\alpha = 0$. Every candidate is feasible. If the query optimum is at most t, its two side lengths occur in this search and their extremal endpoints preserve compatibility; the maximum candidate is therefore optimal. For a longer optimum this routine still returns a feasible chain, possibly shorter. Only two witness handles are retained. Its total overhead is bounded by $O((n+m)t\log^6(n+m+2))$.

5.2 Long answers and final witnesses

For each $z \in R$, compute longest strict paths from z to every reachable northeast endpoint and from every reachable southwest endpoint to z. Use strict-rank sweeps, reversed for backward paths, storing lengths and parent links. Any attaining deterministic tie rule suffices here; these trees need not be the canonical sampling trees. The cost is $O(n\log^2(n+2))$ per z. Put the endpoint coordinates, lengths and handles in static two-dimensional range-max structures.

For a query containing z, maximize forward and backward lengths over endpoints in the query. Both paths lie in the query by endpoint-box containment, and strict index order makes their only common vertex z. Their concatenation is feasible and has length equal to the two lengths minus one. If z lies on a canonical optimum, the maximized halves are at least its corresponding halves. Their feasible concatenation must therefore be optimal. Lemma 5 supplies such a z whenever the optimum is at least t; the short routine handles optima at most t. Take the largest feasible candidate, including the empty one. Threshold equality causes no gap. The stitching overhead is $O((n+m)|R|\log^3(n+m+2))$.

Keep only lengths and constant-size candidate handles during comparison. Finally expand one winner per query. A stitching witness follows its two parent trees, reverses one list in an output-sized buffer, and suppresses the duplicated z. A short witness follows its stored length-state links. Expansion takes $O(K)$ overall; losing candidates are never expanded.

6 Total cost, uniformity and limitations

On the branch $k < N$, the charged overhead is

$$\begin{aligned} O\big((n+m)t\log^6(n+m+2) + Nk\log^3(n+2) \\ + Nk^2\log^2(n+2) + (n+m)k\log^3(n+m+2)\big). \end{aligned} \tag{10}$$

With $t = \Theta(n^{2/3})$, $N < 2n$ and $k = O(n^{1/3}\log^3(n+2))$, this is within Theorem 1. In particular, $O(n^{5/3}\log^8(n+2))$ construction is absorbed by the $(n+m)n^{2/3}$ overhead with its stated logarithms.

On the fallback branch $k \ge N$, take R to be all real points and build no compression tables. All-point stitching costs $O((n+m)n\log^3(n+m+2))$. The branch condition gives $n \le k$, and the formula for k still gives $k = O(n^{1/3}\log^3(n+2))$ on this branch. Thus the same theorem bounds this fallback, without an uncharged finite threshold. The conservative constants can make the fallback apply over a very large finite range. For $n = 1$ direct comparison costs $O(m+1+K)$; if $m = 0$ return immediately.

All sizes, addresses, counters, ranks and finite integer intermediates are polynomial in $n+m$. A fixed number of words accommodates constant-factor larger bit lengths, and the arithmetic charges in Lemma 4 are included in (10). Original real values are only compared and stored. Equal values have equal ranks and never follow each other. Open or closed query boundaries are normalized by the corresponding comparison

searches. The simultaneous hitting property is independent of the number, overlap or density of supplied queries. This completes the proof of Theorem 1 in the full reporting model.

Matching the randomized square-root overhead deterministically is a natural next quantitative goal. Extending the original-interval oracle to other range optimization problems is another direction.

Computational evidence. The accompanying standard-library Python programs check canonical paths, interval traces, rounded probabilities, signed tree sums, merge invariants and reporting on the finite domains documented in the source package. Their reports distinguish compression fixtures with certified initial potential from uncertified fixtures, for which only the recorded algebraic identities are asserted. The uniform correctness and running-time bounds follow from the proof above.

AI methodology. OpenAI Codex was used in problem selection, algorithm and proof development, literature analysis, verification code, and manuscript drafting and compilation. Separate Codex contexts supplied the model-assisted mathematical checks and additional independently written verification programs.

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